

Personalized and Context-Adaptive Conversational AI

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Project Repository & Data Availability: <https://github.com/surnella/QM640-Dialogue-Clustering-Capstone>

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1.1 Background and Problem Domain

Modern conversational artificial intelligence systems are increasingly deployed across high-stakes, consultative enterprise domains such as corporate travel management, healthcare intake, and wealth advisory services (Jannach et al., 2021). Despite significant breakthroughs in large language model (LLM) architectures, state-of-the-art commercial agents suffer from a fundamental behavioral limitation: they rely almost exclusively on static, one-size-fits-all system prompts. Consequently, the agent exhibits rigid, monolithic conversational pacing, identical tonal characteristics, and invariant elicitation strategies regardless of whether the interacting human is an exploratory, hesitant first-time buyer requiring consultative reassurance or a time-sensitive, transactional corporate traveler seeking instantaneous booking confirmation (Inaba et al., 2022). This structural misalignment between user personality archetype and system conversational pacing leads to user friction, repetitive clarifying turns, and premature session abandonment.

Simultaneously, when practitioners attempt to adapt conversational style dynamically through prompt engineering, they encounter severe factual hallucination and policy drift (Rebedea et al., 2023). Injecting diverse, unconstrained conversational exemplars into the LLM context window often causes the model to leak stale constraints, hallucinate pricing or dates from the exemplar dialogue, and violate non-negotiable enterprise operational policies (e.g., booking without valid passport lead time or exceeding discount thresholds). There is an urgent empirical need to formulate a data-driven framework that bridges unsupervised dialogue clustering, dynamic few-shot prompt conditioning, and strict declarative rule enforcement to achieve personalized, style-convergent, yet factually grounded conversational interactions.

1.2 Theoretical Foundation and Prior Art (MultiWOZ 2.2, Frames, S-BERT)

This study builds directly upon foundational literature in dialogue state tracking, sentence representation learning, and in-context learning. The primary theoretical baseline is established by the MultiWOZ benchmark (Budzianowski et al., 2018), specifically the MultiWOZ 2.2 release (Zang et al., 2020), which standardized dialogue act schemas, cleaned state annotation errors, and introduced schema-guided dialogue ontology definitions. To model multi-intent, negotiation-heavy travel interactions where customers dynamically revise constraints across turns, this study integrates the Microsoft Frames corpus (El Asri et al., 2017). Frames formalizes frame tracking, where conversational agents maintain multiple competing candidate itineraries as users explore trade-offs between flight dates, budget limits, and cabin classes.

In the representation domain, we leverage Sentence-BERT (Reimers & Gurevych, 2019) to construct dense semantic trajectory embeddings. Whereas traditional dialogue analysis relies on bag-of-words or turn-level surface lexical matching, S-BERT utilizes siamese and triplet networks to map entire conversational turn sequences into semantically meaningful vector spaces where geometric distance reflects structural conversational behavior. Finally, this research extends the in-context few-shot learning paradigm formalized by Brown et al. (2020) and refined by Liu et al. (2022). Liu et al. demonstrated that LLM downstream task performance is heavily governed by the semantic proximity of retrieved in-context exemplars. We advance this theory by demonstrating that in-context exemplars act as powerful stylistic conditioning priors that steer conversational tone, pacing, and turn economy without requiring parameter fine-tuning.

1.3 Problem Statement (Independent and Dependent Variables)

Existing goal-oriented conversational systems lack data-driven mechanisms to automatically discover latent user conversational archetypes from historical dialogue logs and

adapt their interaction style in real-time, resulting in conversational friction, redundant elicitation, and high abandonment. Furthermore, unconstrained prompt-based style steering introduces factual leakage and policy drift that commercial platforms cannot tolerate.

In a quantitative data analytics framework, this research evaluates how key Independent Variables—specifically, Clustering Algorithm (K-Means vs. Agglomerative vs. HDBSCAN), In-Context Conditioning Strategy (Zero-Shot Baseline vs. Static Few-Shot vs. Dynamic Cluster-Adaptive Few-Shot), Business Rule Enforcement Mode (Rules OFF vs. Declarative Rules Active), and In-Context Exemplar Count (k in $\{1, 3, 5\}$)—statistically impact primary measurable Dependent Variables: Structural Cluster Quality (Silhouette Score and Davies-Bouldin Index), Semantic Style Convergence (Sentence Embedding Cosine Similarity and BERTScore to reference cluster centroids), Factual Slot Accuracy (Joint Goal Accuracy and Token-Level Slot F1), Exemplar Factual Leakage Rate (%), Operational Rule Adherence Rate (RAR, %), and Conversational Efficiency (Turns to Goal Completion [TGC] and Redundant Elicitation Rate [RER]).

1.4 Purpose of the Study

The purpose of this quantitative data analytics study is fourfold: (a) to discover, evaluate, and interpret latent conversational archetypes from human-human travel dialogues using dense trajectory embeddings; (b) to quantify the degree of semantic and stylistic convergence achieved by dynamic cluster-adaptive few-shot prompt injection compared to static baselines; (c) to conduct a controlled 2x2 ablation experiment isolating the causal impact of declarative business rules versus few-shot conditioning on factual slot preservation and policy adherence; and (d) to evaluate conversational efficiency gains across distinct traveler archetypes. The investigation delivers a rigorous, fully reproducible methodology for context-adaptive conversational AI.

1.5 Scope, Boundaries, and 10-Week Feasibility

The scope of this research is strictly bounded to open-access, multi-turn English-language goal-oriented dialogue datasets (primarily Microsoft Frames and MultiWOZ 2.2), benchmarked against an external validation set of 100 sanitized, PII-redacted enterprise golden conversations. The research utilizes dense sentence representation models (all-MiniLM-L6-v2 and BGE-large-en-v1.5) to vectorize entire conversation trajectories. Clustering is restricted to distance-based and density-based algorithms operating in Euclidean and cosine metric spaces. The project is highly feasible within a standard 10-week capstone semester timeline because the entire pipeline is code-driven (Hugging Face Sentence-Transformers -> Scikit-Learn -> FAISS -> Gemini Flash API -> Statsmodels), eliminating manual catalog authoring or complex cloud database hosting. Core data analytics, model execution, and hypothesis testing are completed by Week 8, leaving Weeks 9 and 10 dedicated to documentation beautification, argumentation refinement, and final defense preparation.

1.6 Research Questions (RQ1–RQ4)

To guide the investigation, four focused research questions are established:

- Research Question 1 (RQ1 - Unsupervised Trajectory Clustering Separation): Which unsupervised clustering algorithm (K-Means, Agglomerative Hierarchical, or HDBSCAN) operating on dense trajectory embeddings produces the highest structural cluster separation (Silhouette Score and Davies-Bouldin Index) and distinct behavioral archetypes across multi-turn goal-oriented dialogues?
- Research Question 2 (RQ2 - Dynamic In-Context Semantic Style Convergence): To what extent does dynamic cluster-adaptive few-shot exemplar injection increase the agent's

Semantic Style Closeness (Cosine Similarity and BERTScore to target reference cluster centroids) compared to zero-shot and static few-shot baselines?

- Research Question 3 (RQ3 - Controlled Ablation: Rules vs. Style Factual Drift): In a controlled 2x2 ablation experiment (Declarative Rules [Active vs. OFF] x Prompt Conditioning [Zero-Shot vs. Cluster Few-Shot]), does aggressive stylistic convergence induce Exemplar Factual Leakage or degrade Slot F1, and does a decoupled declarative rule engine eliminate policy violations without dampening style convergence?

- Research Question 4 (RQ4 - Turn Economy & Friction Across Archetypes): How does cluster-adaptive few-shot prompting impact conversational efficiency—measured via Turns to Goal Completion (TGC) and Redundant Elicitation Rate (RER)—when interacting across distinct customer conversational archetypes (e.g., Consultative/Ambiguous vs. Direct/Transactional)?

2. Research Methodology and Sample Size Determination

2.1 Method Selection and Statistical Power Rationale

To ensure robust statistical validity and prevent Type I and Type II errors, sample size calculations were performed for each research question. One-Way Analysis of Variance (ANOVA) power analysis was selected for RQ1 and RQ2 to test differences across clustering algorithms and prompt conditioning treatments. A Two-Way Factorial ANOVA power calculation was utilized for the 2x2 ablation design in RQ3. For RQ4, the Confidence Interval method for population means was applied to estimate dialogue efficiency metrics within a tight margin of error. Across all tests, standard statistical thresholds are enforced: significance level $\alpha = 0.05$ (two-tailed, $Z_{\alpha/2} = 1.96$) and statistical power $1 - \beta = 0.80$ ($Z_{\beta} = 0.8416$).

2.2 Stepwise Sample Size Calculations (RQ1–RQ4)

Stepwise Calculation for RQ1 (One-Way ANOVA - Cluster Quality Differences)

RQ1 compares Silhouette Scores across 3 clustering algorithms (K-Means, Agglomerative, HDBSCAN). To detect a medium standardized effect size of Cohen's $f = 0.25$ at $\alpha = 0.05$ and power = 0.80 across $k = 3$ groups, Cohen's formula is applied:

Degrees of freedom $df_{\text{groups}} = k - 1 = 2$. Critical $F_{(0.05, 2, \infty)} = 3.00$.

Total sample size is given by: $N_{\text{total}} = \lambda / f^2$, where non-centrality parameter $\lambda = 9.64$ for $\alpha=0.05$, power=0.80, $k=3$.

Step 1: Compute f^2 : $(0.25)^2 = 0.0625$.

Step 2: Divide λ by f^2 : $9.64 / 0.0625 = 154.24$.

Step 3: Divide by 3 groups: $154.24 / 3 = 51.41$ per algorithm.

Rounding up yields $n = 52$ dialogue samples per group ($N_{RQ1} = 156$ total cluster evaluation units).

Stepwise Calculation for RQ2 (One-Way ANOVA - Semantic Style Convergence)

RQ2 assesses Semantic Cosine Similarity across 3 conditioning strategies (Zero-Shot, Static Few-Shot, Cluster-Adaptive Few-Shot). To detect a moderate effect size of Cohen's $f = 0.28$ (reflecting anticipated prompt steering differentials) with $\alpha = 0.05$ and power = 0.80 across 3 groups:

Step 1: Compute f^2 : $(0.28)^2 = 0.0784$.

Step 2: Divide $\lambda = 9.64$ by 0.0784: $9.64 / 0.0784 = 122.96$.

Step 3: Divide by 3 groups: $122.96 / 3 = 40.98$ per group.

Rounding up yields $n = 42$ dialogues per treatment arm ($N_{RQ2} = 126$ total dialogue evaluations).

Stepwise Calculation for RQ3 (Two-Way Factorial ANOVA - 2x2 Ablation Experiment)

RQ3 evaluates the 2x2 ablation design (Rules [Active vs. OFF] x Prompting [Zero-Shot vs. Cluster Few-Shot]) on Factual Slot F1 and Rule Adherence Rate. For a 2x2 factorial structure (4 cells), to detect a main effect or interaction effect of size $f = 0.25$ at $\alpha = 0.05$ and power = 0.80:

Degrees of freedom $df_{num} = 1$. Critical $F_{(0.05, 1, inf)} = 3.84$. Required non-centrality $\lambda = 7.85$.

Step 1: Total sample size $N_{total} = \lambda / f^2 = 7.85 / 0.0625 = 125.6$ dialogues.

Step 2: To account for potential dialogue truncation or incomplete synthetic runs, an operational buffer of 40% is incorporated: $125.6 * 1.40 = 175.84$.

Step 3: Allocating equally across 4 factorial cells yields $n = 45$ dialogues per cell.

Total required sample size is $N_{RQ3} = 4 * 45 = 180$ multi-turn dialogues.

Stepwise Calculation for RQ4 (Confidence Interval Method - Turn Economy Estimation)

RQ4 estimates the mean Turns to Goal Completion (TGC) and Redundant Elicitation Rate (RER) within a precise margin of error E . Based on prior dialogue corpus literature (El Asri et al., 2017), dialogue length exhibits standard deviation $\sigma = 3.5$ turns. To achieve a narrow 95% confidence interval margin of error $E = 0.35$ turns (representing 10% of a standard deviation) at $Z_{(\alpha/2)} = 1.96$:

Step 1: Compute numerator: $(Z_{(\alpha/2)} * \sigma)^2 = (1.96 * 3.5)^2 = (6.86)^2 = 47.0596$.

Step 2: Compute denominator: $E^2 = (0.35)^2 = 0.1225$.

Step 3: Divide numerator by denominator: $47.0596 / 0.1225 = 384.16$.

Rounding up yields $N_{RQ4} = 385$ dialogue interactions.

2.3 Sample Size Summary and Final Decision (N = 390)

To ensure all four research questions achieve full statistical validity without underpowering any subgroup analysis, the final sample size decision follows the formal minimax decision rule: $N_{\text{final}} = \max(N_{RQ1}, N_{RQ2}, N_{RQ3}, N_{RQ4}) = \max(156, 126, 180, 385) = 385$ dialogues. To facilitate balanced cross-validation across 3 cluster archetypes and 4 experimental folds, the sample size is rounded to exactly $N = 390$ dialogues (providing 130 dialogues per behavioral archetype). This final decision is highlighted in Table 1 below.

Table 1

Sample Size Determination Summary and Final Minimax Decision across Research Questions

RQ #	Analytical Focus	Method	Alpha	Power	Target f / d	Required N
------	------------------	--------	-------	-------	--------------	------------

RQ1	Cluster Separation (Silhouette)	One-Way ANOVA	0.05	0.80	$f = 0.25$	$N1 = 156$
RQ2	Style Convergence (Cosine Sim)	One-Way ANOVA	0.05	0.80	$f = 0.28$	$N2 = 126$
RQ3	2x2 Ablation (Rules vs Prompting)	2x2 Fact. ANOVA	0.05	0.80	$f = 0.25$	$N3 = 180$
RQ4	Turn Economy (Turns to Goal)	Confidence Int.	0.05	N/A	$E = 0.35, s=3.5$	$N4 = 385$
Final	Comprehensive Study Cohort	Minimax Decision	0.05	≥ 0.80	Max(RQ1-RQ4)	$N = 390$

Note. Final sample size $N = 390$ satisfies the minimax criterion $N = \max(N_{RQ1}..N_{RQ4})$ and enables equal partition across 3 cluster archetypes.

3. Data Infrastructure, Two-Tier Ontology, and Project Repository

3.1 Primary Open Research Datasets and Access Links

To guarantee 100% academic reproducibility, this research utilizes widely recognized, open-access multi-turn conversational corpora. The primary training and clustering dataset is the Microsoft Frames corpus (El Asri et al., 2017), comprising 1,369 human-human dialogues (19,986 individual turns) collected through a Wizard-of-Oz interface. In Frames, users search for vacation packages while exploring complex constraints (dates, destination trade-offs, budget adjustments, and cabin classes). The corpus is available via Microsoft Research at:

<https://www.microsoft.com/en-us/research/project/frames-dataset/>

For schema validation and dialogue state tracking cross-evaluation, we incorporate MultiWOZ 2.2 (Zang et al., 2020), which provides 10,438 multi-domain dialogues spanning hotel, attraction, flight, and taxi domains. MultiWOZ 2.2 features standardized slot definitions and corrected dialogue state labels, available at:

<https://github.com/budzianowski/multiwoz>

3.2 Two-Tiered Canonical Slot Schema (Meta-Ontology Alignment)

A central methodological challenge when combining diverse conversational datasets is ontology divergence. Microsoft Frames uses slot labels such as `or\_city`, `dst\_city`, `str\_date`, and `budget`, whereas MultiWOZ employs `hotel-pricerange`, `flight-arriveBy`, and `taxi-departure`. Furthermore, real-world enterprise operations require operational slots (e.g., passport validity, visa processing windows) absent from academic benchmarks. To resolve this, we adopt the meta-ontology alignment methodology formalized in MultiWOZ 2.2 (Zang et al., 2020) and construct a Two-Tier Canonical Slot Schema:

- Tier 1 (Universal Core Slots): Standardized across all datasets and models, including: Origin_City, Destination_City, Departure_Date, Return_Date, Duration_Nights, Traveler_Count (Adults/Children), Budget_Amount, Accommodation_Tier, and Booking_Status.
- Tier 2 (Enterprise Extension Slots): Domain-specific operational parameters evaluated in RQ3/RQ4, including: Visa_Processing_Window_Days, Passport_Expiry_Months, Package_Type (Group/FIT), Sharing_Basis (Twin/Single), and Discount_Applied_Percentage.

This two-tier schema ensures seamless slot mapping across heterogeneous datasets while isolating enterprise business rules from prompt formatting.

3.3 Cross-Domain Reference Clustering: Public vs. Enterprise Norms

To validate whether dialogue archetypes discovered in public research corpora generalize to real-world corporate travel operations, we employ an external evaluation benchmark of 100 enterprise travel counseling dialogues from an Indian outbound leisure travel agency. To prevent spurious lexical clustering based on entity names (e.g., 'Goa' vs. 'Paris'), all named entities and currency symbols are normalized to generic tokens ('[DESTINATION]', '[ORIGIN]', '[CURRENCY]') prior to embedding. Enterprise conversation trajectories are projected into the latent vector space established by the public Frames corpus, enabling direct measurement of cross-domain centroid alignment.

3.4 Ethical Considerations and Algorithmic Bias

Because conversational AI agents can inadvertently amplify demographic stereotypes or exhibit exclusionary behavior, rigorous ethical safeguards are enforced. First, privacy and confidentiality are preserved: the public datasets are anonymized, and the 100 enterprise benchmark conversations undergo automated Named Entity Recognition (NER) scrubbing with

regex filtering to purge names, phone numbers, credit card tokens, and employee IDs (complying with FERPA and NDA standards). Second, algorithmic bias in few-shot conditioning is actively mitigated. Cluster archetypes are defined purely on conversational mechanics (e.g., interaction length, turn latency, constraint revision frequency) rather than user demographic attributes, ensuring equitable agent behavior.

3.5 Project GitHub Repository, Folder Structure, and Replication Instructions

To fulfill the 2.5-point rubric requirement for a working, well-organized data repository, a dedicated public GitHub repository has been established at:

<https://github.com/surnella/QM640-Dialogue-Clustering-Capstone>

The repository (is currently empty, but will be complete in the following weeks) is structured to ensure complete end-to-end reproducibility, organized into the following directory tree:

```

QM640-Dialogue-Clustering-Capstone/
├── data/
│   ├── raw/                # Ingested Microsoft Frames and MultiWOZ 2.2 JSON files
│   ├── canonical/          # Preprocessed Two-Tier mapped dialogues
│   ├── embeddings/         # Cached S-BERT 384-dimensional trajectory embeddings
│   └── enterprise_sample/   # 100 PII-sanitized evaluation dialogues
├── notebooks/
│   ├── 01_edata_cleaning.ipynb    # Exploratory data analysis & cleaning logs
│   ├── 02_trajectory_clustering_rq1.ipynb # RQ1 K-Means, Agglomerative, HDBSCAN
│   ├── 03_fewshot_retrieval_rq2.ipynb  # RQ2 FAISS indexing & Gemini inference
│   ├── 04_factorial_ablation_rq3.ipynb  # RQ3 2x2 Factorial ANOVA (Rules vs Drift)
│   └── 05_turn_economy_rq4.ipynb       # RQ4 Turn economy & friction statistics
├── src/
│   ├── ingestion.py           # MultiWOZ and Frames JSON parsers
│   ├── canonical_schema.py    # Pydantic schemas for Tier-1 and Tier-2 slots
│   ├── vectorizer.py          # Turn-level and trajectory Sentence-Transformers
│   ├── clustering.py          # Scikit-learn & HDBSCAN clustering pipelines
│   ├── faiss_index.py         # Exact L2/Cosine vector search for exemplar retrieval
│   ├── prompt_assembler.py   # Dynamic few-shot prompt construction
│   ├── rule_engine.py         # Decoupled declarative business rule validator
│   └── evaluation.py          # Automated evaluators (BERTScore, Slot F1, Leakage)
├── rules/
│   └── rules_enterprise.yaml   # Declarative business constraints (visa, pax, budget)
├── outputs/
│   ├── figures/               # UMAP projections, Silhouette plots, interaction charts
│   └── tables/                # Exported ANOVA summary tables and CSV logs
├── requirements.txt           # Pinned Python package dependencies
└── README.md                 # Replication guide, Docker setup, and documentation

```

4. Comprehensive Data Dictionary

Table 2 provides the complete, tabular data dictionary defining all experimental variables, operational measures, data types, units, and methodological roles required for evaluating RQ1–RQ4.

Table 2

Comprehensive Experimental Data Dictionary for Goal-Oriented Dialogue Clustering and Adaptive Prompting

Variable Name	Data Type	Role	Operational Definition & Meaning	Units / Range	Relevance
dialogue_id	String	Identifier	Unique dialogue session code from Frames/MultiWOZ	Discrete ID	Unit of analysis for trajectory tracking
clustering_algo	Categorical	Independent	Algorithm: K-Means, Agglomerative, HDBSCAN	3 Categories	Primary factor for RQ1 cluster evaluation
conditioning_mode	Categorical	Independent	Prompting mode: Zero-Shot, Static Few-Shot, Cluster Few-Shot	3 Categories	Treatment factor for RQ2 style convergence
rule_engine_mode	Categorical	Independent	Ablation condition: 0=Rules OFF, 1=Declarative Rules Active	Binary {0, 1}	Treatment factor for RQ3 ablation experiment
target_archetype	Categorical	Independent	Assigned dialogue cluster (Explorer, Negotiator, Transactional)	3 Clusters	Stratification factor across RQ2–RQ4
exemplar_count_k	Integer	Independent	Number of in-context demonstration dialogues injected	Count (k in {1, 3, 5})	Evaluates demonstration density effects
silhouette_score	Float	Dependent	Measure of cluster cohesion and separation	Score [-1.0, 1.0]	Primary metric for RQ1 evaluation
davies_bouldin_idx	Float	Dependent	Ratio of within-cluster distance to separation	Index [0.0, inf)	Validation metric for RQ1 evaluation
cosine_similarity	Float	Dependent	Cosine similarity between generated turn and cluster centroid	Score [-1.0, 1.0]	Primary style convergence

					metric for RQ2
bert_score_f1	Float	Dependent	Token-level semantic similarity to target reference exemplars	Score [0.0, 1.0]	Secondary semantic alignment metric for RQ2
slot_f1_score	Float	Dependent	Harmonic mean of slot extraction precision and recall	Ratio [0.0, 1.0]	Factual slot preservation metric for RQ3
factual_leakage_rate	Float	Dependent	Proportion of hallucinated entities traced to exemplar prompt	Ratio [0.0, 1.0]	Direct measure of prompt contamination in RQ3
rule_adherence_rate	Float	Dependent	Proportion of turns respecting visa, pax, and blackout rules	Ratio [0.0, 1.0]	Primary compliance metric for RQ3 ablation
turns_to_goal	Integer	Dependent	Total turns required to resolve user goal successfully	Turns (1–25)	Primary conversational efficiency metric for RQ4
redundant_elicit_rate	Float	Dependent	Proportion of turns re-asking established constraints	Ratio [0.0, 1.0]	Dialogue friction metric for RQ4
budget_gap_closure	Float	Dependent	Proportion of initial budget discrepancy resolved	Ratio [0.0, 1.0]	Commercial negotiation metric for RQ4

Note. Variables span trajectory embeddings, prompt conditioning modes, ablation indicators, and evaluation metrics.

5. Analytic Approach and Solution to RQ1–RQ4

This section maps each of the four research questions to its rigorous analytic methodology, detailed technical feature mechanics, mathematical formulations, underlying assumptions, and expected empirical outputs. To ensure complete alignment with committee requirements, all open methodological decisions are formally documented as explicit empirical assumptions.

5.1 Solution to RQ1: Unsupervised Trajectory Clustering Quality

5.1.1 Trajectory Construction Mechanics and Mathematical Formulation

A foundational contribution of this research is the transformation of multi-turn conversational interactions into dense structural trajectory vectors. Let a completed dialogue session D of length T turns be represented as an ordered sequence of user and assistant turn pairs: $D = \{(u_1, a_1), (u_2, a_2), \dots, (u_T, a_T)\}$. Converting D into a fixed-length representation suitable for distance-based clustering requires capturing both semantic evolution and interaction structure without losing temporal progression.

First, each individual turn pair t is mapped into a $d = 384$ -dimensional dense semantic vector using Sentence-BERT (all-MiniLM-L6-v2): $e_t = \text{S-BERT}(u_t \parallel [\text{SEP}] \parallel a_t)$ in $\mathbb{R}^{384}$. Second, because early turns typically define constraint discovery while terminal turns capture negotiation closure or resolution, an exponential temporal decay weighting function w_t is applied across turns:

$$w_t = \exp(\lambda * t / T) / \sum_{k=1}^T \exp(\lambda * k / T)$$

where $\lambda = 0.75$ is the temporal progression parameter governing trajectory sensitivity.

Third, to capture non-lexical conversational mechanics (pacing, hesitation, negotiation friction),

an auxiliary structural feature vector s_D in R^6 is computed:

$s_D = [\ln(T), L_user_avg, L_asst_avg, \Delta_budget_mention, Turn_to_First_Offer, Constraint_Revision_Count]$

where L represents character length and Δ indicates pricing adjustment count. The final unified trajectory vector v_D is constructed via concatenation and L2-normalization:

$v_D = L2_Normalize([\sum_{t=1}^T w_t * e_t] \parallel StandardScaler(s_D))$ in R^{390} .

This trajectory vector captures the joint semantic-structural arc of the dialogue, enabling clustering algorithms to separate interactions based on customer behavioral pacing.

5.1.2 Hypotheses, Analytic Method, and Methodological Assumptions

Hypotheses:

- Null Hypothesis H_0 : There is no significant difference in structural cluster separation (Silhouette Score) across K-Means, Agglomerative Hierarchical, and HDBSCAN algorithms ($\mu_K = \mu_{Agg} = \mu_{HDB}$).
- Alternative Hypothesis H_1 : Significant differences exist in structural cluster separation across algorithms, with density-based or agglomerative methods outperforming standard spherical K-Means ($\mu_{HDB} \neq \mu_K$).

Analytic Method: A One-Way Analysis of Variance (ANOVA) followed by Tukey's Honestly Significant Difference (HSD) post-hoc test evaluates Silhouette Scores across 10 repeated 5-fold cross-validation bootstrap iterations ($N = 156$ evaluation units). Formulae: The Silhouette Score for trajectory i is: $s(i) = (b(i) - a(i)) / \max(a(i), b(i))$, where $a(i)$ is mean intra-cluster distance and $b(i)$ is mean nearest-cluster distance. The Davies-Bouldin Index is: $DB = (1/k) * \sum_{i=1}^k \max_{j \neq i} ((\sigma_i + \sigma_j) / d(c_i, c_j))$.

Methodological Assumptions (Addressing Committee Inquiries):

1. Normality & Homogeneity: Bootstrap Silhouette distributions satisfy normality and homoscedasticity (Levene's test $p > .05$).
2. Universal Geometric Latent Space: It is assumed that latent conversational trajectory structures discovered in large public corpora (Microsoft Frames and MultiWOZ 2.2) represent universal human goal-oriented negotiation arcs (Hesitant Exploration, Budget Friction, Rapid Transactional). Consequently, private enterprise travel conversations can be projected directly into this public embedding space via the identical S-BERT vectorizer without retraining embeddings, satisfying corporate privacy constraints while benchmarking cross-domain validity. Expected Outputs: Identification of optimal $k = 3$ behavioral archetypes (Hesitant Explorer, Budget Negotiator, Rapid Transactional) with Silhouette Score ≥ 0.42 and Davies-Bouldin Index ≤ 1.15 .

5.2 Solution to RQ2: Dynamic In-Context Few-Shot Style Convergence

5.2.1 Exemplar Selection Logic, Centroid Triggers, and Vector Retrieval

To operationalize adaptive prompt steering during live multi-turn interactions, a precise exemplar retrieval trigger and matching pipeline is established. Exemplar selection operates via a two-stage real-time trigger mechanism:

1. Dynamic Retrieval Trigger: The exemplar selection mechanism is triggered at session initialization (turn $t = 1$) using the user's opening intent, and is re-evaluated at turn $t = 2$ upon receipt of the first constraint preference expression. The cumulative ongoing dialogue prefix $P_t = \{(u_1, a_1), \dots, u_t\}$ is vectorized in real time: $q_t = \text{S-BERT}(\sum_{i=1}^t w_i * \text{Turn}_i)$.
2. Centroid Proximity Matching: Query vector q_t is compared against precomputed cluster centroids $\{C_1, C_2, \dots, C_K\}$ using cosine similarity: $k^* = \arg\max_{j \in \{1..K\}} ((q_t \cdot C_j) / (\|q_t\| \|C_j\|))$. If max cosine similarity exceeds confidence threshold $\tau_{\text{conf}} \geq 0.65$, the

interaction is assigned to target archetype k^* .

3. FAISS Sub-Index Nearest-Neighbor Query: Within the isolated FAISS vector sub-index

$I_{\{k^*\}}$, an exact inner-product search retrieves the top $k = 3$ most structurally similar historical golden dialogues: $E = \text{Top-3}_{\{d \in I_{\{k^*\}}\}} ((q_t \cdot d) / (\|q_t\| \|d\|))$.

4. Slot-Masked Prompt Formatting: The retrieved golden dialogues undergo entity de-lexicalization (replacing specific dates, cities, and prices with canonical slot tokens) before being formatted into the Gemini Flash API system prompt prefix. This ensures the LLM absorbs the conversational style, tone, and pacing of archetype k^* without contaminating current dialogue state with stale entity values.

5.2.2 Hypotheses, Analytic Method, and Methodological Assumptions

Hypotheses:

- Null Hypothesis H_0 : Dynamic cluster-adaptive few-shot conditioning does not significantly increase Semantic Cosine Similarity to the target reference cluster compared to static few-shot prompting ($\mu_{\text{cluster}} = \mu_{\text{static}}$).
- Alternative Hypothesis H_1 : Dynamic cluster-adaptive few-shot conditioning produces significantly higher Semantic Cosine Similarity and BERTScore to target reference cluster centroids ($\mu_{\text{cluster}} > \mu_{\text{static}}$).

Analytic Method: A One-Way Repeated Measures ANOVA compares Cosine Similarity across three prompting treatments (Zero-Shot Baseline, Static Few-Shot, Dynamic Cluster Few-Shot) evaluated across $N = 126$ dialogue sessions (42 per condition). Formulae: Cosine Similarity between generated system response trajectory g and target cluster centroid c is: $\text{Cos_Sim} = (g \cdot c) / (\|g\| * \|c\|)$. BERTScore computes token-level greedy matching cosine similarity using contextualized RoBERTa embeddings.

Methodological Assumptions (Addressing Committee Inquiries):

1. Sphericity & Independence: Sphericity is evaluated via Mauchly's test; if violated, Greenhouse-Geisser epsilon adjustments are applied.
2. Stylistic Prior Independence: It is assumed that in-context few-shot exemplars act purely as stylistic priors that guide turn pacing and language formality, while slot values and factual commitments remain under the control of the dialogue state tracker and business rule engine.

Expected Outputs: Demonstration of statistically significant style convergence (Cosine Similarity increase from 0.58 baseline to ≥ 0.82 under dynamic conditioning, $p < .001$, Cohen's $d \geq 0.85$).

5.3 Solution to RQ3: Controlled 2x2 Factorial Ablation of Rules vs. Style Factual Drift

5.3.1 Deterministic Goal Completion Criteria and Multi-Factor Success Rule

To rigorously quantify whether conversational style convergence introduces factual hallucination or compromises task success, a deterministic, unambiguous goal completion rule is enforced. A dialogue session D is classified as a binary success ($Y_{\text{success}} = 1$) if and only if all three of the following Boolean criteria are simultaneously satisfied:

Criterion 1: Mandatory Slot Saturation: All Tier-1 mandatory core slots (Origin_City, Destination_City, Departure_Date, Duration_Nights, Traveler_Count, Budget_Tier) must be fully populated in the final dialogue state tracker: $\text{State_slots} \cap \text{Required_slots} = \text{Required_slots}$.

Criterion 2: Zero Unresolved Policy Violations: Automated programmatic evaluation against the declarative rule schema yields zero blocking violations ($V_{\text{blocking}} = 0$), verifying that no package was offered violating visa lead times, exceeding customer budget caps, or breaching passenger capacity limits.

Criterion 3: Explicit Terminal Commitment Act: The assistant successfully executes a formal commitment action (Offer_Package or Confirm_Itinerary) and the user responds with an explicit acceptance dialogue act (Inform(Accept) or Affirm()), concluding before max turns $T \leq 20$.

Mathematically: $\text{Success}(D) = I((\text{Slot_Saturation} == \text{True}) \text{ and } (\text{V_blocking} == 0) \text{ and } (\text{Terminal_Acceptance} == \text{True}))$. This deterministic definition eliminates subjective evaluation bias.

5.3.2 Hypotheses, Analytic Method, and Methodological Assumptions

Hypotheses:

- Null Hypothesis H0: Active declarative rules do not significantly improve Rule Adherence Rate (RAR) or prevent factual leakage under cluster conditioning (Main Effect $\beta_{\text{rules}} = 0$).
- Alternative Hypothesis H1: Active declarative rules significantly increase Rule Adherence Rate ($\text{RAR} \geq 0.98$) and eliminate factual leakage, proving business rules causally enforce policy compliance without dampening style convergence.

Analytic Method: A Two-Way Factorial ANOVA (2x2 design: Declarative Rules [Active vs. OFF] x Prompt Conditioning [Zero-Shot vs. Cluster Few-Shot]) evaluates Slot F1, Exemplar Leakage Rate, and Rule Adherence across $N = 180$ dialogues (45 per cell). Formulae: Exemplar Factual Leakage Rate is: $\text{Leakage} = (\text{Hallucinated Entities Traced to Prompt Exemplars}) / (\text{Total Generated Entities})$. Rule Adherence Rate is: $\text{RAR} = 1 - (\text{Violations} / \text{System Turns})$.

Methodological Assumptions (Addressing Committee Inquiries):

1. Decoupled Orthogonality Assumption: It is assumed that enterprise business rules can be completely decoupled into an external declarative configuration (rules_enterprise.yaml) executed outside the LLM context window. The 2x2 factorial ANOVA assumes no unmodeled confounding between prompt structure and rule execution, allowing the main effect of rules and

the interaction effect (Rules x Prompting) to isolate causal policy protection.

2. Error Distribution: Factorial residuals meet assumptions of normality and homoscedasticity across the four balanced cells.

Expected Outputs: Proof that declarative rules eliminate factual leakage (reducing leakage from ~14.2% in Rules-OFF down to 0.0% in Rules-Active) while maintaining high Slot F1 (≥ 0.92).

5.4 Solution to RQ4: Dialogue Turn Economy and Friction Analysis

5.4.1 Declarative Rule Schema Scope and Precise Boundary Conditions

To establish reproducible boundaries for conversational efficiency and compliance, the declarative rule engine (`rules_enterprise.yaml`) enforces four explicit classes of operational boundary conditions validated deterministically across all interactions:

1. Temporal and Visa Feasibility Boundaries:

- Minimum Booking Lead Time: International departures require a minimum lead time of ≥ 21 calendar days when visa processing is pending; domestic itineraries require ≥ 48 hours.

- Passport Expiry Threshold: International bookings require passport validity of ≥ 6 months beyond the scheduled return date.

2. Financial and Commercial Boundaries:

- Maximum Discount Ceiling: Negotiated promotional discounts cannot exceed 10.0% of catalog base fare without human manager escalation.

- Hard Budget Cap Adherence: If a customer specifies a strict upper budget ceiling $B_{\max}$, the agent is strictly prohibited from proposing an itinerary where $\text{Price} > B_{\max}$.

3. Capacity and Operational Boundaries:

- Minor Accompaniment: Child travelers (< 18 years) require at least one adult passenger; infant-to-adult ratio must not exceed 1.0.

- Hotel Room Occupancy: Standard double occupancy enforces a maximum of 3 adults per room (requiring an extra rollaway bed).

4. Conversational and Legal Compliance Boundaries:

- Mandatory Branding Greeting: Opening turn must include organizational identity and agent greeting.
- Mandatory Disclosure of Terms: Cancellation policy and travel insurance disclaimers must be presented prior to terminal booking confirmation.

5.4.2 Hypotheses, Analytic Method, and Methodological Assumptions

Hypotheses:

- Null Hypothesis H0: Cluster-adaptive prompting does not significantly reduce Turns to Goal Completion (TGC) or Redundant Elicitation Rate (RER) across archetypes ($\mu_{\text{cluster}} = \mu_{\text{baseline}}$).
- Alternative Hypothesis H1: Cluster-adaptive prompting significantly reduces TGC and RER, with the greatest turn reductions observed in complex consultative archetypes ($\mu_{\text{cluster}} < \mu_{\text{baseline}}$).

Analytic Method: A Two-Way Factorial ANOVA evaluating Turns to Goal Completion across Conditioning Mode (3 levels) and Customer Archetype (3 levels) across $N = 385$ dialogue interactions. Formulae: Redundant Elicitation Rate is defined as: $\text{RER} = (\text{Redundant Constraint Questions}) / (\text{Total System Turns})$, where a question is redundant if the requested slot is already saturated in state.

Methodological Assumptions (Addressing Committee Inquiries):

1. Two-Tier Ontology Invariance: It is assumed that the Two-Tier Canonical Slot Schema harmonizes dataset schema differences without information loss, allowing turn economy metrics

to be compared fairly across public and enterprise dialogue sessions.

2. Distributional Characteristics: Dialogue turn counts approximate a log-normal distribution;

log-transformed turn counts are utilized in ANOVA models to guarantee normality.

Expected Outputs: Significant reduction in dialogue turns (18–25% reduction in TGC for consultative users) and reduction in RER from 0.16 down to ≤ 0.03 .

6. Full Solution Pipeline and System Architecture

The proposed experimental system implements an automated, reproducible machine learning and NLP pipeline structured into four distinct architectural modules:

Module 1: Trajectory Embedding & Unsupervised Clustering: Historical dialogues from Microsoft Frames are preprocessed, canonicalized to Tier-1/Tier-2 slots, and vectorized using Sentence-Transformers (all-MiniLM-L6-v2) combined with structural features into 390-dimensional trajectory vectors. Clustering algorithms (K-Means, HDBSCAN) partition trajectories into structural behavioral archetypes (Explorer, Negotiator, Transactional). Clusters are indexed in a local FAISS vector database.

Module 2: Dynamic Few-Shot Nearest-Neighbor Retrieval: When an incoming customer inquiry turn is received, its embedding is computed in real-time. FAISS queries the target cluster index to retrieve the top $k = 3$ most structurally relevant demonstration dialogues, which undergo slot masking.

Module 3: Decoupled Dual Compliance & Prompt Assembler: The retrieved exemplars are formatted into a structured prompt prefix. Crucially, business rules are maintained in a decoupled declarative YAML file (`rules_enterprise.yaml`) and evaluated before and after LLM generation via deterministic assertion wrappers, ensuring rules and prompts operate independently.

Module 4: Automated Evaluation Harness: Automated evaluators compute: (a) Cosine Similarity and BERTScore to reference centroids; (b) Token-level Slot F1 against MultiWOZ ground truth; (c) Exemplar Factual Leakage Detection via named entity recognition cross-referencing; and (d) Conversational turn efficiency metrics logged to SQLite.

7. 10-Week Phased Project Implementation Plan

To fulfill Walsh College QM640 rubric requirements and align precisely with the official course schedule (Schedule- QM 640 - Sheet1), the project is organized across a realistic 10-week implementation lifecycle. In strict accordance with course milestone policies, all core data analytics, model engineering, and hypothesis testing are completed by Week 8 ('Data Analysis Assessment'). Week 9 ('Data Analysis Refinement and Documentation') and Week 10 ('Document research results and finding') are dedicated to documentation beautification, slide deck finalization, and formal capstone defense preparation. Week 11 provides post-submission refinement for minor committee adjustments with zero new analytical scope. Table 3 maps the official QM640 syllabus topics directly to our project milestones, analytical tasks, and deliverables.

Table 3

Ten-Week Phased Project Implementation Schedule, Deliverables, and QM640 Syllabus Milestone Alignment

Week & Dates	QM 640 Syllabus Topic	Capstone Phase & Analytical Tasks (Dialogue Clustering)	Tools & Stack	Course Deliverable & Status
Week 0 (Sep 26)	Introduction to Capstone	Problem statement selection and finalization; evaluate travel conversational AI vs clustering thesis	Git, Markdown, Python 3.11	Problem Statement Finalized
Week 1 (Sep 27 - Oct 3)	Define Research Problem and Topic	Scaffold GitHub repository; ingest raw Microsoft Frames & MultiWOZ 2.2 corpora; establish environment	Python, Pandas, GitHub, JSON	GitHub Repository Scaffolded
Week 2 (Oct 4 - Oct 10)	Synopsis Draft Submission	Formalize RQ1-RQ4; complete Two-Tier slot schema mapping; submit Synopsis Draft (Oct 6); live review (Oct 10)	Docx, Statsmodels, SciPy	Synopsis Draft Submitted (Oct 6)
Week 3 (Oct 11 - Oct 17)	Literature Search and Data Collection	Synopsis Final Submission (Oct 13); extract S-BERT 384d trajectory embeddings; generate EDA cleaning logs	S-BERT, Hugging Face, UMAP	Final Synopsis & Vectorized Corpus
Week 4 (Oct 18 - Oct 24)	Formulate Objectives & Hypotheses	Interim Report Draft Submission (Oct 20); train baseline K-Means, Agglomerative,	Scikit-Learn, HDBSCAN,	Interim Report Draft (Oct 20)

HDBSCAN (RQ1); live review (Oct 24)			Seaborn	
Week 5 (Oct 25 - Oct 31)	Choose Research Design & Methodology	Interim Report Final Submission (Oct 27); construct FAISS sub-indexes for top-k exemplar retrieval; validate clusters	FAISS-CPU, Pydantic, YAML	Final Interim Report (Oct 27)
Week 6 (Nov 1 - Nov 7)	Data Analysis Plan	Final Report Draft Submission (Nov 3); implement Gemini Flash inference pipeline across Zero-Shot vs Dynamic Few-Shot (RQ2)	google-genai SDK, Gemini Flash	Final Report Draft & Generated Logs
Week 7 (Nov 8 - Nov 14)	Data Analysis Plan Execution	Execute 2x2 Factorial Ablation (Rules ON/OFF x Prompting); evaluate Slot F1, factual leakage, and RAR (RQ3)	Statsmodels, PyYAML, SQLite	RQ3 Ablation Matrix Completed
Week 8 (Nov 15 - Nov 21)	Data Analysis Assessment	Complete Turn Economy (TGC) & Redundant Elicitation (RER) tests (RQ4); execute all One-Way/Two-Way ANOVAs; ALL ANALYTICS DONE	Statsmodels, Pingouin, Matplotlib	Data Analysis 100% Completed
Week 9 (Nov 22 - Nov 28)	Data Analysis Refinement & Documentation	Capstone Final Presentation (Nov 27/28); Final Presentation Submission (Nov 29); documentation beautification & slide deck	PowerPoint, LaTeX, Seaborn	Final Presentation Slide Deck
Week 10 (Nov 29 - Dec 6)	Document Research Results and Finding	Author 15-page comprehensive capstone thesis; finalize GitHub documentation; Final Report Submission (Dec 13)	Word, GitHub, Zenodo	Final Capstone Thesis (Dec 13)
Week 11 (Dec 7 - Dec 13)	Refine Documentation of Results & Finding	Post-submission committee refinements; incorporate faculty feedback; final project archive (no new analytical scope)	GitHub Releases, PDF Archive	Final Archive Completed

Note. Implementation schedule perfectly matches the official QM640 schedule. All computational and statistical data analysis is 100% completed by Week 8 (Data Analysis Assessment), reserving Weeks 9–10 for documentation refinement, presentation delivery, and thesis defense.

8. Practical Recommendations, Applications, and Stakeholder Value

8.1 Target Users and Beneficiaries

Beneficiaries span three key stakeholder groups: (1) Conversational AI Product Teams and AI Engineers, who gain an automated, data-driven method for persona steering without manual prompt re-writing; (2) Customer Experience and CRM Operations Leaders, who can deploy bots that naturally mirror diverse customer pacing and tones, boosting engagement; and (3) NLP Researchers, who receive the first systematic empirical characterization of the trade-off between in-context stylistic convergence and factual hallucination leakage.

8.2 Actionable Recommendations and Business Impact

Organizations deploying LLM conversational agents should transition from static prompt engineering to Dynamic In-Context Clustering. Specifically: (a) cluster historical enterprise conversation logs to discover natural customer archetypes; (b) implement vector similarity nearest-neighbor lookup to supply style exemplars on-the-fly; and (c) enforce strict decoupled declarative rule engines to prevent exemplar factual leakage. In enterprise settings, dynamic persona matching is projected to increase dialogue task completion by 20–30% while eliminating policy violations.

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